Bugs found: 6

1. Out-of-bounds pointer arithmetic: 'q = p + 15' points outside the allocated 10-element array.
2. Out-of-bounds write: '*q = 42' writes to memory outside the allocated buffer via invalid pointer q.
3. Use-after-free read: '*p' is dereferenced after 'free(p)', resulting in undefined behavior.
4. Memory leak risk / dangling pointer: 'p' is not set to NULL after free, leaving a dangling pointer.
5. Buffer overflow: loop writes 6 characters into 'str[5]', exceeding the array bounds by one element.
6. Missing null terminator: 'str' is never null-terminated, causing 'printf("%s")' to invoke undefined behavior.